

# JEE MAIN REVISION MASTER: CHEMICAL EQUILIBRIUM

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 118 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                            | Frequency | Category        |
|---------------|-----------------------------------------|-----------|-----------------|
| Very High     | Chemical Equilibrium - Core Application | 118 PYQs  | General Concept |

## 2. Master Formula Reference Card

| Concept / Topic | Mathematical Formulation & Governing Equations |
|-----------------|------------------------------------------------|
|-----------------|------------------------------------------------|

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Chemical Equilibrium - Core Application (Very High Yield — 118 PYQs)

**Core Principle:** Key problem pattern in Chemical Equilibrium.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

**Representative Source PYQs:**

- [2026 Evening Shift] Q#1 (MCQ): "Consider the following reactions in which all the reactants and products are present in gaseous state  $\begin{matrix} \text{A} + 2\text{B} \rightleftharpoons \text{C} + 2\text{D} \end{matrix}$  ... " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Observe the following equilibrium in a 1 L flask.  $\text{A(g)} \rightleftharpoons \text{B(g)}$  At T(K), the equilibrium concentrations of A and B are 0.5 M and 0.375 M respectively." [Ans: N/A]
- [2026 Evening Shift] Q#1 (Numerical): "In a closed flask at 600 K, one mole of  $\text{X}_{(2)}\text{Y}_{(4)}(\text{g})$  attains equilibrium as given below :  $\text{X}_{(2)}\text{Y}_{(4)}(\text{g}) \rightleftharpoons 2\text{XY}_{(2)}(\text{g})$  ..." [Ans: 8]

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

In a closed flask at 600 K, one mole of  $\text{X}_{(2)}\text{Y}_{(4)}(\text{g})$  attains equilibrium as given below :  $\text{X}_{(2)}\text{Y}_{(4)}(\text{g}) \rightleftharpoons 2\text{XY}_{(2)}(\text{g})$  At equilibrium, 75%  $\text{X}_{(2)}\text{Y}_{(4)}(\text{g})$  was dissociated and the total pressure is 1 atm. The magnitude of  $\Delta_r G^\circ$  (in  $\text{kJ mol}^{-1}$ ) at this temperature is \_\_\_\_\_. (Nearest Integer) (Given :  $R=8.3\text{J mol}^{-1}\text{K}^{-1}$ ;  $\ln 10=2.3$ ,  $\log 2=0.3$ ,  $\log 3=0.48$ ,  $\log 5=0.69$ ,  $\log 7=0.84$ )

**Correct Answer:** 8

**Solution Analysis:**  $P^\circ(1+\alpha)=1$   $\alpha=(3/4)$ ;  $P^\circ=(4/7)$   $K_{\text{eq}}(P)=\frac{P_{\text{XY}}^2}{P_{\text{XY}_2}}=\frac{(3/7)^2}{(4/7)}=(3/4)$   $\Delta_r G^\circ=-RT \ln K_{\text{eq}}(P)$   $|\Delta_r G^\circ|=8.314 \times 600 \ln((36/7))=8169.5\text{J mol}^{-1}=8.169\text{kJ mol}^{-1}$